Assessing uncertainty in DNA evidence caused by sampling effects

JM Curran*
Department of Statistics, University of Waikato, Hamilton, New Zealand
JS Buckleton
Forensic Science Service, Trident Court, Solihull Parkway, Birmingham Business Park, Solihull B37 7YN, United Kingdom
CM Triggs
Department of Statistics, University of Auckland, Auckland, New Zealand
BS Weir
Program in Statistical Genetics, North Carolina State University, Raleigh, NC 27695-8203, USA
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Sampling error estimation in forensic DNA testimony was discussed. Is an estimate necessary and how should it be made? The authors find that all modern methods have areas of strength and weakness. The assessment of which is the 'best' is subjective and depends on the performance of the method, the type of problem (criminal work or paternity), the database size and availability of computing software and support. The authors preferred the highest posterior density approach for performance, however the other methods all have areas where their performance is adequate. For single-contributor stains normal approximation methods are suitable, also the bootstrap and the highest posterior density method. For multiple-contributor stains or other complex situations the match probability expressions become quite complex and it may not be possible to derive the necessary variance expressions. The highest posterior density or the bootstrap provide a better general method, with non-zero θ. The size-bias correction and the factor of 10 approaches may be considered acceptable by many forensic scientists as long as their limitations are understood.

L'estimation de l'erreur de l'échantillonage dans le témoignage de l'ADN forensique est discutée. Une estimation est-elle nécessaire et comment doit-elle être faite? Les auteurs trouvent que toutes les méthodes modernes présentent des zones de force et de faiblesse. L'évaluation de ce qui est le meilleur est subjectif et dépend de la performance de la méthode, du type de problème (affaire criminelle ou en paternité), de la base de données et de la disponibilité de programmes informatiques et de soutien. Les auteurs préférent l'approche de la densité postérieure la plus élevée, pour la performance toutefois, les autres méthodes ont toutes des domaines où leurs performances sont adéquates. Pour des taches d'un seul contributeur, les méthodes d'approximation normales sont adéquates, tout comme le bootstrap et la méthode de la densité postérieure la plus élevée. Pour des taches de contributeurs multiples et d'autres situations complexes les expressions de probabilité de concordance deviennent tout à fait complexes et il se peut qu'il ne soit pas possible de dériver des expressions de la variance nécessaire. La densité postérieure la plus élevée, ou le bootstrap donne une méthode générale meilleure avec un θ non nul. La correction par le biais dû à la taille et le facteur de dix approches peuvent être considérés comme acceptables par bien des scientifiques pour autant que leur limitation soient comprises.

Es wird die Abschätzung von Probenahmefehler in forensischen DNA-Zeugenaussage diskutiert. Ist eine Abschätzung notwendig und wie sollte diese gemacht werden? Nach Ansicht der Autoren weisen alle modernen Methoden Stärken und Schwächen auf. Die Bewertung, welche der Methoden die beste ist, ist subjektiv und hängt von ihrer Leistungsfähigkeit, von der Art der Problemstellung (kriminaltechnische Untersuchung oder Vaterschaftsuntersuchung), von der Datenbankgröße sowie von der Verfügbarkeit der Computersoftware ab. Obwohl andere Modelle Bereiche haben, in denen sie eine adäquate Leistungsfähigkeit besitzen, bevorzugen die Autoren die Methode der "Highest Posterior Density". Für Spuren von Einzelpersonen sind normale Approximationsmethoden sowie auch der "Bootstrap" und die "Highest Posterior Density"-Methode geeignet. Für Mischspuren oder andere komplexe Situationen kann die Formel für die Trefferwahrscheinlichkeit so komplex werden, dass man u.U. die notwendigen Varianzen nicht ableiten kann. Die "Highest Posterior Density" oder der "Bootstrap" eignen sich mit $\theta \neq 0$ besser als generelle Methode. Die "size-bias Korrektur" und die "Factor of 10"-Methode dürfen von vielen forensischen Wissenschaftlern als akzeptable erachtet werden, solange ihre Nachteile verstanden werden.

Se discute la estimación del error de muestreo en el testimonio forense de ADN. Es necesaria esta estimación? y como debe hacerse? Los autores encuentran que todos los métodos modernos tienen puntos débiles y puntos fuertes. La valoración de cual es la mejor manera de hacerla es subjetiva y depende de la ejecución del método, el tipo de problema (trabajo criminal o paternidad), el tamaño de la base de datos y la disponibilidad de ordenadores y soporte informático. Los autores prefirieron para la ejecución, el método de la máxima densidad a posteriori aunque los otros métodos tienen todos áreas donde la aplicación es adecuada. Para las manchas de un solo individuo, los métodos de aproximación son adecuados, también la bootstrap y el método de la máxima densidad a posteriori. Para las manchas de varios individuos u otras situaciones complejas la expresiones de la probabilidad de coincidencia son bastante complejas y puede resultar imposible deducir de ellas las necesarias expresiones de varianza. El método de la máxima densidad a posteriori o el de bootstrap proporcionan mejor método general con una θ no –zero. La corrección del sesgo por el tamaño de la muestra y los métodos del factor 10 pueden considerarse aceptables para muchos expertos forenses con tal de que se conozcan sus limitaciones.

*** Author for correspondence**
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Introduction

DNA profiling evidence has gained a deserved position as one of the most powerful forensic tools of modern times. Typical cases may include those where a suspect has the same profile as a stain at the crime scene or has possessions stained with a victim's profile. In such cases it is usual to attach numerical weight to the event of the observed match to aid interpretation by investigators or triers of fact in court [1]. These weights are generally based on frequencies of the allelic components of the profile. These frequencies are estimated from profiles collected for differing purposes and from databases which may be quite modest in size. As such the frequencies are sample estimates and are subject to statistical sampling error. There are at least two questions of forensic interest in this context: should an estimate of this sampling uncertainty be made? and how should this be done?

When scientists or informed members of the public use statistics, they generally use summary measures such as the mean or the median. The idea is to describe the 'average' or 'central' value. However there are many contexts where the mean or median of a population does not constitute an adequate summary. In such cases the mean may be inadequate for a person to make a reasoned decision about a particular matter. To focus this discussion we take up an example given by Gould [2]. Consider a headline that states that: 'Under the current government mean take-home pay has risen 20%'. This may be an impressive fact, one sufficiently impressive to make many people vote for this government again. However, the earning population may be separated into two categories: the 99% who used to earn $10 a year but now earn $9 and the remaining 1% who used to earn $1 million dollars but now earn $1,200,297 a year. It is true that the mean earnings have risen 20% but most people are worse off. The information in the statement about the mean gives a completely false impression and may lead to different voting decisions at election time.

Could this be the situation in DNA evidence? The answer is: 'Yes but probably to a much lesser extent.' The example given above is related to a highly skewed distribution where the mean gave a misleading impression. It is believed that the sampling distribution for match probability estimates is also skewed but to a lesser extent [3]. Even in the case where the distribution is not skewed it would still be more scientifically 'complete' to give an assessment of uncertainty.

We can certainly imagine some cases where the evidence, DNA and other, is so overwhelming that no realistic adjustment for uncertainty in the match probability or likelihood ratio will have any effect on the outcome of the case. However there may be other cases where this is not so. In any case we feel that it is incumbent on a forensic witness to convey a reasonably complete understanding. In our view this involves more than merely giving a mean estimate. It should include an assessment of the uncertainty in the estimate. Furthermore, the relevancy or appropriateness of these issues is not for the forensic witness to decide.

Scientists should find it interesting that this point needs to be argued at all. It is almost universally accepted that any measurement in science is associated with some measure of uncertainty. Why should DNA evidence be exempt? However an estimate of uncertainty from sampling effects is not universally made. What is it that exempts DNA evidence from what is standard scientific and statistical practice?

The National Research Council report on 'The evaluation of Forensic DNA evidence' [4] devotes a chapter to 'Confidence intervals for match probabilities' where they describe confidence intervals as 'helpful.' We would agree with this but go further and suggest that they are necessary to provide a full understanding of the evidence.

In summary we suggest that an assessment of sampling error is necessary in presenting DNA evidence and that most modern commentators support this position [1, 3–8]. We therefore take up the matter of which of the offered methods is most suitable for forensic purposes.

Methods for assessing uncertainty from sampling effects

In this section we assume that we have convinced the reader to assess uncertainty in the match probability estimate caused by sampling effects. Convinced readers would obviously be interested in an effective, easily implemented and scientifically sound way to do this.

All methods for assessing sampling error require a certain amount of input. Initially one must decide on the type of interval (one or two-sided) and the desired level of confidence (a discussion of Bayesian posterior density intervals follows later). A two-sided, 99%, confidence interval would allow the following type of statement to be made:

The 99% confidence limits for the match probability are 1 in a million and 1 in 35 million. 99% of confidence intervals constructed in this way should contain the true value.

In conventional frequentist statistics a confidence interval is interpreted in terms of repeated sampling. The match probability is calculated from a database containing a fixed number of profiles, say N, from a specified population. Thus if a very large number of databases each containing N profiles were sampled from the population, and a match probability and confidence interval calculated using the allele frequencies from each database, then the intervals from 99% of databases would include the true match probability for the population as a whole. The one-sided equivalent is:

The 99% one-sided confidence limit is 1 in 1.1 million. In 99% of such intervals the true frequency should be less than this value.

Note that the limit changes slightly and that the one-sided confidence limit has one bound rather than two. Any particular interval will either contain the true match probability or it will not, thus in strict terms 99% confidence is not equivalent to a

probability of 0.99. Thus it is not strictly correct to substitute the word *probability* for the word *confidence* in these descriptions of confidence intervals. We discuss later a Bayesian approach that allows more straightforward statements to be made. In many practical situations the Bayesian posterior density intervals and frequentist confidence intervals are numerically very similar, despite the difference in interpretation between the Bayesian and frequentist approaches.

There have been several methods suggested to characterise the sampling error of an estimated match probability, generally through the use of confidence intervals. The first NRC forensic DNA report suggested that the frequency of each constituent allele be replaced by a binomial upper confidence limit, and these limits then multiplied across alleles [5 p.92].

Another simplistic approach, although one with some empirical support, was suggested by the second NRC forensic DNA report and was to multiply and divide the estimated match probability by a factor of 10 [4 p.149]. Weir suggested a bootstrapping approach by resampling individuals in some population sample [9]. Chakraborty et al appealed to the asymptotic normality of the logarithm of the product of allele frequencies to provide an analytical expression for the confidence interval [8], and this was extended by the National Research Council [4 p.147] to allow for population structure. Balding gave both a likelihood support interval and a 'conservative Bayesian estimate' [3].

We will describe each of these methods in some detail and compare their performances by simulation. We will also present a new Bayesian method that makes use of the posterior distribution of the match probability.

Match probabilities

DNA profiles consist of genotypes at l loci. At the *ith* locus two alleles A_{i1}, A_{i2} are observed. These alleles may be the same in which case the genotype is described as a homozygote at this locus, or different in which case the genotype is described as a heterozygote. We write the probability of obtaining allele A_{ij} in some population as p_{ij}. Match probabilities, P_i, for locus i [1, 10] are

$$P_i=\begin{cases}\dfrac{[3\theta+(1-\theta)p_{i1}][2\theta+(1-\theta)p_{i1}]}{(1+\theta)(1+2\theta)}, & A_{i1}=A_{i2}\\ \dfrac{2[\theta+(1-\theta)p_{i1}][\theta+(1-\theta)p_{i2}]}{(1+\theta)(1+2\theta)}, & A_{i1}\neq A_{i2}\end{cases} \quad (1)$$

These are the probabilities of a person in the population having genotype $A_{i1}A_{i2}$ given that one person in the population is known to have the genotype and are the terms required in most forensic applications. The quantity θ, known as the coancestry coefficient, also written as F_{ST}, describes the structure of the population. In particular it allows use of population-wide allele probabilities p_{ij} in the absence of knowledge of the probabilities in the specific subpopulation relevant to the crime in question. When this substructure is either absent or is ignored, θ can be set to zero, and (1) can be simplified to

$$P_i=\begin{cases}p_{i1}^2, & A_{i1}=A_{i2}\\ 2p_{i1}p_{i2}, & A_{i1}\neq A_{i2}\end{cases} \quad (2)$$

This last expression is known as the 'product rule.' Whether or not θ is set to zero, it is customary to multiply over loci to get the overall match probability P

$$P=\prod_i P_i$$

In place of the unknown population probabilities p_{ij}, it is usual to use the allele proportions $\hat{p}_{ij}$ obtained from a sample of n_i individuals from the population (not from the specific relevant subpopulation) that were scored for locus i. The corresponding estimated match probabilities will be written as $\hat{P}_i$ and $\hat{P}$. We wish to assess the sampling error associated with $\hat{P}$.

The second NRC report (NRC II) makes two suggestions. We discuss both of these here.

The factor of 10

The suggestion under discussion here is to construct the interval $(\hat{P}/10, 10\hat{P})$ about the estimate. We begin by discussing whether NRC II actually meant this seriously as a recommendation. It appears on p.160 under the heading 'Conclusions and Recommendations.' This strongly suggests that they 'recommend' this approach. However the idea does not receive a 'Numbered recommendation'. This seems to indicate that it should not be taken as seriously as the 'numbered' recommendations. The relevant paragraph from the NRC II is as follows:

'The empirical studies show that the differences between the frequencies of the individual profiles estimated by the product rule from adequate subpopulation databases, (at least several hundred persons) are within a factor of about 10 of each other, and that provides a guide to the uncertainty of the determination for a single profile.'

At the very least, this approach is easily implemented and has some empirical support, however, it is lacking in some other respects. The general meaning of the paragraph quoted above is clear but it contains a number of words that are vague and may lead to unhelpful argument in court, viz., adequate, subpopulation, several, about, and guide. Perhaps the weakest word is 'several' which would have different meanings for different people.

The extent of the uncertainty must depend on, at least, the number of loci, the sample size, and the nature of the estimator whose uncertainty is being assessed. Small databases should lead to higher uncertainty in the estimate. The level of uncertainty will increase with increasing numbers of loci. Some estimators have an intrinsic insensitivity to sampling uncertainty. These statements should be self explanatory except the last. For example the variability in the formulae for brothers (2a) should be more insensitive to sampling variation than equation (1) because they are dominated by constant terms (the fixed leading terms of $^1/_4$ in the formulae below).

$$P_i = \begin{cases} \dfrac{(1+p_{i1})^2}{4}, & A_{i1} = A_{i2} \\ \dfrac{1+p_{i1}+p_{i2}+2p_{i1}p_{i2}}{4}, & A_{i1} \neq A_{i2} \end{cases} \quad (2a)$$

The 'factor of 10' method is a very coarse tool, however it does have some empirical support and, in fact, this simulation will provide some further support.

Normal distribution approach

Methods based on asymptotic normality have the advantage of leading to relatively simple analytical expressions. The logarithm of the estimate $\ln(\hat{P})$ is the sum of l independent estimates $\ln(\hat{P}_i)$ and so is approximately normally distributed. Hence a 99% confidence interval for P is given by $(\hat{P}/C, C\hat{P})$ where $\ln(C)=2.57\sqrt{\mathrm{Var}[\ln(\hat{P})]}$. The variance of $\ln(\hat{P})$ which has been given as

$$\mathrm{Var}[\ln(\hat{P}_i)] = \begin{cases} \dfrac{2(1-\hat{p}_{i1})}{n_i\hat{p}_{i1}}, & A_{i1} = A_{i2} \\ \dfrac{\hat{p}_{i1}+\hat{p}_{i2}-4\hat{p}_{i1}\hat{p}_{i2}}{2n_i p_{i1}p_{i2}}, & A_{i1} \neq A_{i2} \end{cases}$$

([8]; [4], p.146), (however this requires that $\theta = 0$.)

This approach is easily implemented. For instance it can be relatively easily set up in an EXCEL™ spreadsheet for, say, the CODIS core loci, and would allow the practicing forensic scientist to calculate a one or two sided confidence interval for $\hat{P}$ using either the product rule or equation 1.

Bootstrap procedure

The bootstrap procedure discussed by Weir [9] resamples profiles from the set of profiles determined for individuals sampled from the population, i.e. the database. The idea is that by resampling profiles, rather than alleles, any within-individual allelic dependencies in the sample are preserved. For each bootstrap sample, allele proportions at all loci are found and the match probability estimated. The resulting estimated match probabilities are ordered and a $100(1-\alpha)\%$ confidence interval is created from these values. A two-sided confidence interval is created by taking the central $100(1-\alpha)\%$ of these values. An upper $100(1-\alpha)\%$ confidence interval for frequency is taken by taking the upper $100(1-\alpha)\%$.

The bootstrap procedure requires a purpose written program. However it does have the advantage of being easily extended to deal with mixtures or paternities.

Support intervals

Suppose that the population probability of allele j at locus i is p_{ij}, and that a sample from the population contains x_{ij} copies of that allele. For locus i, where the sample has n_i scored individuals, Balding [3] proposed the following support intervals:

$$P_{0i} = \begin{cases} \dfrac{(x_{i1}-2\lambda)^2}{(2n_i+2\lambda)^2}, & A_{i1} = A_{i2} \\ \dfrac{2(x_{i1}+\lambda)(x_{i1}+\lambda)}{(2n_i+2\lambda)^2}, & A_{i1} \neq A_{i2} \end{cases} \quad (3)$$

These formulae are appealing because of a supposed natural interpretation, namely that the parameter λ corresponds to the addition of λ copies of the profile to the database. If we choose $\lambda = 2$, then (3) can be thought of as 'the effect of adding the suspect's profile and the crime scene profile to the database'. Balding based his ideas on a profile likelihood interval, a likelihood analogue to a confidence interval.

A 99% profile likelihood interval for the profile frequency is found by choosing two values of λ that give the correct properties. Unfortunately in most cases the λ value for the upper support limit should not be two. The bounds of λ will depend on the number of loci in the profile, the population probabilities of the profile alleles, and the size of the sample as well as on the probability ordinate used to construct the interval. Adding the crime stain profile and the suspect's profile to the database before constructing allele sample frequencies neither accounts for the sampling variation induced by finite databases, nor corrects for the presence of population substructure.

In addition there appears to be no simple general extension of these formulae to mixed stains cases and paternity cases.

Prior distributions for allele frequencies

Balding ([3] corrected in [1]) also considered a Bayesian approach by assuming a Dirichlet prior distribution for the probabilities of the alleles observed in the profile. The posterior probability of the profile without accounting for substructure is:

$$\frac{E(P_{0i}^2)}{E(P_{0i})} = \begin{cases} \dfrac{(x_{i1}+4)(x_{i1}+3)}{(2n_i+5)(2n_i+4)}, & A_{i1} = A_{i2} \\ \dfrac{2(x_{i1}+2)(x_{i2}+2)}{(2n_i+6)(2n_i+5)}, & A_{i1} \neq A_{i2} \end{cases} \quad (4)$$

It has been implemented in an approximate form:

$$\frac{E(P_{0i}^2)}{E(P_{0i})} = \begin{cases} \dfrac{(x_{i1}+4)(x_{i1}+4)}{(2n_i+4)(2n_i+4)}, & A_{i1} = A_{i2} \\ \dfrac{2(x_{i1}+2)(x_{i2}+2)}{(2n_i+4)(2n_i+4)}, & A_{i1} \neq A_{i2} \end{cases}$$

which represents the effect of adding the profile in question to the database exactly twice and has been termed 'the size bias correction'.

Such estimates have been referred to as 'conservative' Bayesian estimates but this is difficult to justify. They are approximately posterior means for match probabilities.

This approach remains the accepted standard in some laboratories. We assume that the appeal, again, is simplicity and ease of implementation.

Highest posterior density

It is relatively easy to refine Balding's method to produce the complete posterior distribution and indeed, Balding has set out the method for this. This involves calculation of the complete posterior distribution, assuming again a uniform prior. However, rather than propose a uniform prior for the frequency of the unknown alleles in question (say, jointly uniform for the frequencies of allele A and not for a homozygote, and jointly uniform for the three frequencies A, B and not A or B for a heterozygote), a joint uniform prior is proposed for the frequency of all of the alleles at the locus. This has the effect of making prior expectation the same regardless of whether one considers a homozygous or heterozygous profile. There is no requirement to assume a uniform prior and other priors may be used. In many cases the posterior is insensitive to the prior. Once the complete distribution has been calculated the $100(1-\alpha)\%$ highest posterior density interval (one or two-sided) may be chosen.

Simulation results

The theoretical considerations discussed above may have less importance to the practising forensic scientist than practical considerations such as: 'How well do these methods work?' The need here is to see how the method to be assessed performs in terms of 'capturing' the true value. It would be preferable to trial these methods in casework and to see how often they make a true statement about the, say, US population. Unfortunately the true value of, say, the frequency of a certain profile in people other than the suspect is typically not available. Resort is therefore made here to computer simulation in which the 'true values' are known. We intend to 'score' the various methods and it is necessary to set up a mechanism to do so. We imagine a linear scale of, say, frequency with the true value existing somewhere along it. We initially consider a two-sided confidence interval. It seems reasonable to feel that one that captured the true value between narrow bounds would be particularly useful whereas one that did not capture the true value, especially if it missed it by overestimating rarity, would be wrong. A method that did capture the true value but between particularly wide bounds (such as zero and one for a probability) would be strictly correct but of less use.

When we shift these considerations to a one-sided confidence interval for, say, frequency, we see that we want the majority of our one sided limits to lie just to the right of the true value. Those well to the right of the true value are excessively conservative (too large), whereas those to the left give an incorrect answer (too small).

Simulation description

Use was made of the New Zealand Caucasian database of size n=1,764. The database contains profiles using the six loci: vWA, HUMTHO1, D8S1179, FGA, D21S11 and D18S51. Apart from HUMTH01, these loci are a subset of the 13 core loci used by US forensic agencies [11,12] and the African American ($n = 210$) database published by Budowle et al [11,12]. Every profile in these databases was assigned a match probability (assuming independence between and within loci). Using this information a common profile (with a relatively large match probability) and a rare profile (with a very small match probability) were selected. The performance of each method was tested using these 'common' and 'rare' profiles. More common or more rare profiles could have been artificially generated.

The sample allele proportions from the databases were used to generate 1,000 'databases' of n profiles for $n = \{100, 400, 1000\}$. This simulates the act of sampling from the population. As the databases were generated without any population substructure, the appropriate estimator of the match probability is given by the product rule.

Each of the methods outlined above was used to get an estimate of sampling error in the estimated match probabilities for the common and rare profiles. For example, on the first iteration, a database of 100 six locus or thirteen locus profiles was generated. That database was then summarised by determining the frequency of the alleles appearing in our preselected common and rare profiles. These frequencies were then used to calculate the normal approximation. The allelic counts were used to calculate the size bias correction and to generate posterior allele distributions for the Bayesian posterior method. The profiles were resampled for the bootstrap. Therefore, after one iteration we have an estimate of sampling error for each of the four methods of interest. The normal approximation was used to give a 99% upper confidence limit. One thousand (1,000) bootstrap samples were taken from each database for the bootstrap method and the 99th percentile recorded. One thousand samples from the posterior density of the allele probabilities were taken for the highest posterior density (HPD) method: the upper bound of a one-sided 99% confidence/highest posterior density interval for the profile probability was calculated. Balding's Bayesian method was used with Equations (4) (instead of his Equation 12) and a uniform prior (Figures 1–4). The dashed vertical lines in those figures are at the true value of $\log_{10}(P)$ where P is the profile frequency, and should cut off the lower 1% of the curves. Summary statistics are given in Tables 1–4. The percentage of zeros records the percentage of sample estimates that were zero – that is one or more alleles in the profile did not appear in the database and hence the estimated match probability is zero. The coverage probability records the probability that the upper bound of a one-sided 99% confidence/highest posterior density interval exceeded the true value of $\log_{10}(P)$ where P is the profile frequency. This figure should be close to 0.99 (or 99%) as this was the desired significance level. The columns labelled 2.5% and 97.5% give the 2.5th and 97.5th percentiles of the the upper bound of a one-sided 99% confidence/highest posterior density intervals. These can be used as a measure of the variability in each of the methods. The median is the median upper bound of the upper bound of a one-sided 99% confidence/highest posterior density interval calculated by each method. The ratio is the ratio of the median upper bound to the the true value. If the ratio has a value

of ten for example, this means that the median upper bound calculated by the method is ten times higher than the true value.

Each bound has a frequentist interpretation for the normal approximation, bootstrap and support interval, namely 'on average, 99% of the intervals are expected to contain the true population profile frequency'. This is what is implied by the statement, 'We are 99% confident that the interval contains the true population profile frequency'. Correspondingly, the interpretation for the Bayesian HPD is that '99% of the probability density is contained with this interval'.

Discussion

If DNA evidence is to be accompanied by estimated match probabilities, a prudent statistical procedure is to provide a measure of sampling error for these estimates. This is especially necessary when estimates are based on samples of just a few hundred profiles. The criticism that this provision of both an estimate and its bound(s) may confuse a jury can be met if the forensic scientist presents only the upper (or lower) bound(s) and states that sampling effects have been considered.

The most complete discussion of sampling error is based on the sampling distribution of an estimate. This leads naturally to frequentist confidence and support intervals or Bayesian highest posterior density intervals, even if these are described only in terms of their boundaries.

During this simulation the profile estimate was occasionally zero, especially for the rare profile in the small database. We assume that in these circumstances the situation will be 'rescued' by a 'default minimum' allele frequency. However we point out that this is necessarily arbitrary and the performance of the minima needs to be assessed separately to this study. Such corrections are not necessary for the size bias method or the Bayesian posterior method as they assign non-zero prior probability to observing an allele in a database.

For the common profile we score the factor of ten to be too conservative (too far to the right) for most database sizes. Balding's method, the size bias correction, appears to be too far to the left (unconservative) to be accommodating sampling variation in any meaningful way, except for small databases or rare profiles. We score the Bayesian Highest Posterior Density method as too far to the left (non-conservative) side for the common CODIS profile and a database of size 100.

It may be a valid criticism to say 'who cares' for the rare profile since it is already rare and a few powers of 10 may have little effect. We assume for this discussion that the readers are interested in being as correct as possible in their initial estimation. Any rounding or assessments of 'importance of accuracy' can be left for later or, better, to the court. For the rare profile the factor of 10 is variable, being too far to the left occasionally (CODIS n = 100 and n = 400) and once too far to the right (NZ Caucasians n = 1,000). We also score poor performances for Balding's method (NZ Caucasians n = 1,000 too far left, CODIS n = 100 too far to the right), the log normal method (NZ Caucasians and CODIS n = 100 too far to the right).

The relatively poor performance of normal-based limits for the rare profile and small sample sizes is a consequence of specific alleles not appearing in these samples. The problem disappears when θ is assigned a non-zero value.

Both bootstrapping and posterior distribution methods have substantial computing requirements. This becomes noticeably more serious for the bootstrap as the sample size increases but not for the Bayesian posterior. This suggests use of normal-based analytical methods or the Bayesian posterior, especially for large sample sizes. For single-contributor stains, such as those considered here, it does appear that these normal methods are completely suitable. For multiple-contributor stains, however, or other complex situations involving relatives or missing persons [1] the match probability expressions become quite complex and it may not be simple or possible to derive the necessary variance expressions. The highest posterior density or the bootstrap would better provide a general approach, with non-zero θ.

In view of the general interest in the 'Factor of 10' approach and the 'Size Bias Correction' we give these two special consideration.

The 'Factor of 10' approach has little to recommend it from a theoretical standpoint, and we must prefer the other methods which have a more firm statistical backing. As mentioned previously there must be an effect of sample size and number of loci. We are disappointed in the coarseness of this approach as a statistical tool, and point out that it is inappropriate for mixtures, relatives, paternity or missing persons. However, our own simulation may be viewed as further empirical support for the 'Factor of 10' approach. In general the 'Factor of 10' approach performed in a broadly similar way to the other methods or was excessively conservative. On two occasions both involving the rare profile we scored it as inadequately conservative, however this may be of reduced importance in a rare profile. This may be an acceptable situation for practicing forensic scientists.

The 'Size Bias Correction' received some criticism in this paper. However for small databases or rare profiles it is probably acceptable as a coarse correction for sampling error, and this paper could be taken as empirical support for its usefulness in these situations. For larger databases and common profiles it performs more as a mean estimate.

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Table 1 Summary measures for a common profile using NZ Caucasians.

Method	% Zero	Coverage	2.5%	Median	97.5%	Ratio
		***n* = 100**				
Highest Posterior Density	0	0.86	-7.8	-7.3	-6.9	9.2
Bootstrap	0	0.95	-7.7	-7.1	-6.6	8.7
Log Normal	0	0.98	-7.5	-7.0	-6.6	8.4
Balding	0	0.79	-7.9	-7.4	-6.9	9.4
Factor of 10	0	1.00	-7.2	-6.6	-6.1	7.5
		***n* = 400**				
Highest Posterior Density	0	0.95	-7.6	-7.4	-7.1	9.4
Bootstrap	0	0.98	-7.5	-7.3	-7.1	9.3
Log Normal	0	0.99	-7.5	-7.3	-7.0	9.2
Balding	0	0.64	-7.8	-7.5	-7.3	9.9
Factor of 10	0	1.00	-6.8	-6.6	-6.3	7.4
		***n* = 1,000**				
Highest Posterior Density	0	0.96	-7.6	-7.4	-7.3	9.6
Bootstrap	0	0.98	-7.6	-7.4	-7.2	9.5
Log Normal	0	0.98	-7.5	-7.4	-7.2	9.5
Balding	0	0.57	-7.7	-7.5	-7.4	10.0
Factor of 10	0	1.00	-6.7	-6.6	-6.4	7.4

Table 2 Summary measures for a rare profile using NZ Caucasians. *These figures are calculated using non-zero profiles only, and hence may be biased.

Method	% Zero	Coverage*	2.5%*	Median*	97.5%*	Ratio*
		***n* = 100**				
Highest Posterior Density	0.0	1.00	-12.6	-11.9	-11.9	7.8
Bootstrap	75.8	1.00	-12.7	-7.1	-11.9	7.9
Log Normal	75.8	1.00	-12.0	-7.0	-11.3	7.1
Balding	0.0	1.00	-12.4	-7.4	-11.8	7.7
Factor of 10	75.8	1.00	-12.5	-6.6	-11.9	7.9
		***n* = 400**				
Highest Posterior Density	0.0	0.99	-13.1	-12.5	-12.0	8.7
Bootstrap	11.8	0.96	-13.3	-12.7	-12.1	9.0
Log Normal	11.8	1.00	-12.9	-12.4	-11.9	8.5
Balding	0.0	0.96	-13.4	-12.8	-12.2	9.1
Factor of 10	11.8	0.98	-13.2	-12.4	-11.7	8.6
		***n* = 1,000**				
Highest Posterior Density	0.0	0.98	-13.2	-12.7	-12.4	9.1
Bootstrap	0.5	0.95	-13.4	-12.9	-12.4	9.3
Log Normal	0.5	0.99	-13.2	-12.7	-12.3	9.1
Balding	0.0	0.84	-13.6	-13.1	-12.7	9.6
Factor of 10	0.5	1.00	-13.0	-12.3	-11.9	8.5

Table 3 Summary measures for a common profile using CODIS African Americans.

Method	% Zero	*n* = 100 Coverage	2.5%	Median	97.5%	Ratio
Highest Posterior Density	0	0.78	-19.1	-18.2	-17.5	9.6
Bootstrap	0	0.93	-18.8	-17.9	-17.1	9.2
Log Normal	0	0.98	-18.5	-17.7	-16.9	9.0
Balding	0	0.92	-18.8	-18.0	-17.3	9.3
Factor of 10	0	0.97	-18.6	-17.7	-16.8	9.0
		***n* = 400**				
Highest Posterior Density	0	0.93	-18.6	-18.2	-17.8	9.6
Bootstrap	0	0.97	-18.6	-18.1	-17.7	9.5
Log Normal	0	0.98	-18.5	-18.1	-17.7	9.4
Balding	0	0.76	-18.8	-18.4	-18.0	9.8
Factor of 10	0	1.00	-18.0	-17.6	-17.2	8.9
		***n* = 1,000**				
Highest Posterior Density	0	0.97	-18.5	-18.3	-18.0	9.7
Bootstrap	0	0.99	-18.5	-18.3	-18.0	9.6
Log Normal	0	0.99	-18.5	-18.2	-18.0	9.6
Balding	0	0.68	-18.7	-18.5	-18.2	9.9
Factor of 10	0	1.00	-17.8	-17.6	-17.3	8.9

Table 4 Summary measures for a rare profile using CODIS African Americans. *These figures are calculated using non-zero profiles only, and hence may be biased.

Method	% Zero	*n* = 100 Coverage*	2.5%*	Median*	97.5%*	Ratio*
Highest Posterior Density	0.0	1.00	-28.6	-27.1	-25.8	8.4
Bootstrap	76.2	0.99	-29.1	-27.5	-26.0	8.6
Log Normal	76.2	1.00	-27.7	-26.3	-25.1	7.9
Balding	0.0	1.00	-27.1	-26.0	-25.0	7.7
Factor of 10	76.2	0.97	-29.5	-28.0	-26.7	8.9
		***n* = 400**				
Highest Posterior Density	0	1.00	-29.1	-28.1	-27.1	9.0
Bootstrap	12.2	0.97	-29.5	-28.4	-27.4	9.2
Log Normal	12.2	1.00	-28.7	-27.9	-27.0	8.8
Balding	0.0	1.00	-29.0	-28.1	-27.3	9.0
Factor of 10	12.2	0.89	-29.9	-28.6	-27.5	9.4
		***n* = 1,000**				
Highest Posterior Density	0.0	0.99	-29.3	-28.5	-27.9	9.3
Bootstrap	0.4	0.94	-29.7	-28.7	-28.0	9.5
Log Normal	0.4	1.00	-29.1	-28.5	-27.9	9.3
Balding	0.0	0.95	-29.5	-28.8	-28.2	9.5
Factor of 10	0.4	0.93	-29.7	-28.5	-27.7	9.3

Figure 1 Distributions of 99% upper bounds for a common profile using NZ Caucasians.

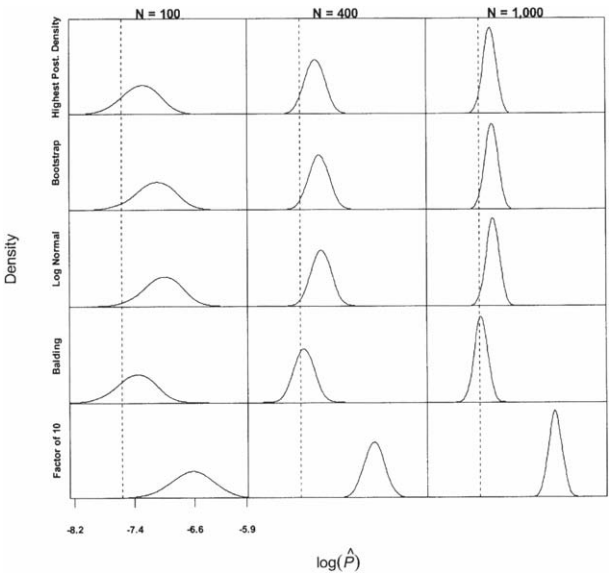


Figure 2 Distributions of 99% upper bounds for a rare profile using NZ Caucasians.

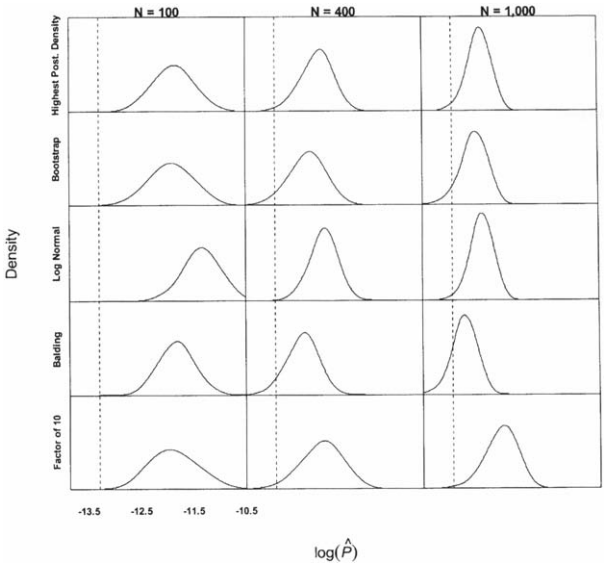


Figure 3 Distributions of 99% upper bounds for a common profile using CODIS African Americans.

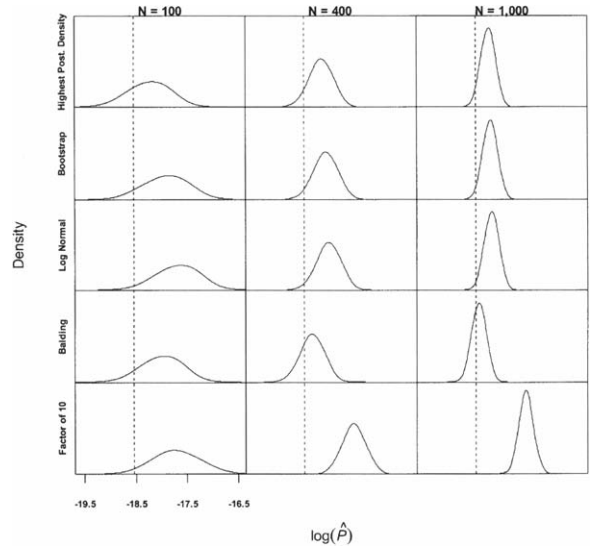
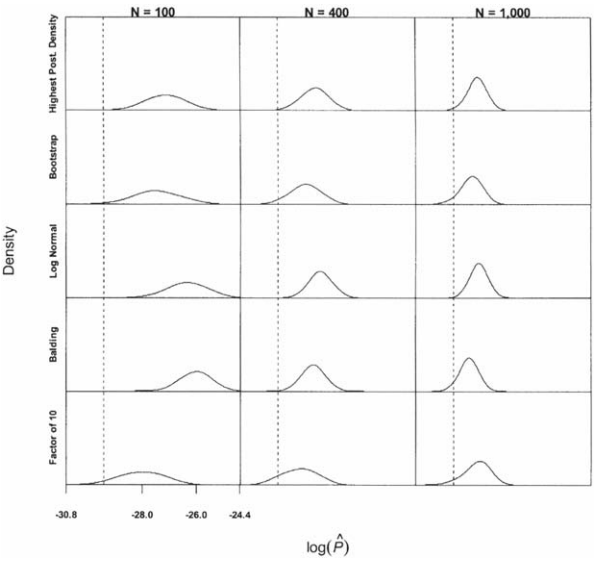


Figure 4 Distributions of 99% upper bounds for a rare profile using CODIS African Americans.



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